Sarah Wheeler

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Education

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
B.S. Information Sciences + Data Science — Expected May 2027
GPA: 4.0/4.0

Research Interests

Human–AI Interaction; Responsible AI; Human Behavior & Decision-Making

Research Experience

Undergraduate Research Assistant — November 2024 – Present
Adaptive Cognition & Interaction Design (ACTION) Lab — University of Illinois Urbana-Champaign

PI: Dr. Jessie Chin

- Investigate human–AI interaction and AI-supported behavior change through *MintBot*, a motivational interviewing chatbot evaluated using a Wizard-of-Oz study design, conducting mixed-methods analysis of interaction data to examine user acceptance, behavioral responses, communication patterns, and intervention efficacy
- Develop a multi-agent AI learning system integrating LLM-based dialogue, belief modeling, and retrieval, with specialized agents coordinating peer discussion, evidence search, and adaptive intervention
- Develop and evaluate an LLM-based belief extraction pipeline that tracks users' evolving factual and preference beliefs from dialogue and identifies misconceptions, knowledge gaps, confidence, and intervention needs
- Analyze 60+ interview transcripts using computational linguistic methods, including Coh-Metrix and NLTK, alongside qualitative thematic coding to identify patterns in users' goals, barriers, and responses to AI-supported behavior change
- Contribute to manuscript development through statistical analysis, data visualization, and results writing on health behavior change, goals, and barriers

Undergraduate Research Assistant — September 2025 – Present
Information Technology Ethics and Accessibility (iTEA) Lab — The University of Texas at San Antonio

PI: Dr. Kyrie Zhixuan Zhou

- Lead a first-author mixed-methods study (N = 366) examining how people negotiate agency, trust, and human–AI collaboration when delegating everyday decisions and tasks to AI systems
- Design the study methodology and survey instrument, prepare IRB materials, and develop measures of trust, privacy, autonomy, economic benefit, and human–AI collaboration
- Develop a quantitative analysis pipeline incorporating correlation analysis, predictive modeling, clustering, and descriptive statistics to identify patterns in attitudes toward AI delegation and human oversight
- Co-author a mixed-methods study of AI-assisted programming, analyzing LLM interaction logs and survey responses to investigate over-reliance and human oversight in AI-supported coding

Manuscripts Under Review

[1] G. Tsai, S. Wakoyi, S. Mangal, R. Deguzman, **S. Wheeler**, J. Shen, & K. Z. Zhou. "Beyond Vibe Coding: How Structured AI Collaboration Supports Programming Learning." *Under review at iConference 2027.*

[2] **S. Wheeler**, V. Rammouz, Y. Qian, S. Shetty, Z. Liu, M. Wei, A. Rios, T. He, & K. Z. Zhou. "What to Delegate, What to Preserve: Human Preferences for Agentic AI Across Everyday Life and Work." *Under review at CSCW 2027.*

[1] M. Bak, **S. Wheeler**, J. Campbell, V. Sridharan, M. S. Imam, S. Bhat, C. Chiu, & J. Chin. "Have a Brief Chat: Designing Motivational Interviewing Chatbots (MintBot) to Support Healthy Choices Over Time." *Under review at IEEE ICHI 2027.*

Manuscripts in Preparation

B. Choi, **S. Wheeler**, B. Ho, & J.Chin. "Understanding Digital Health Information-Seeking Behavior and Lifestyle Change." *In preparation for JMIR.*

Conference & Poster Presentations

[1] **S. Wheeler**, M. Bak, J. Campbell, V. Sridharan, M. S. Imam, S. Bhat, C. Chiu, & J. Chin. "Have a Brief Chat: Examining the Acceptance and Efficacy of Motivational Interviewing Chatbots (MintBot) on Prompting Healthy Choices Over Time." *Illinois Undergraduate Research Symposium*, University of Illinois Urbana-Champaign, 2026.

Outstanding Undergraduate Poster Award

Policy & Public Scholarship

[1] M. Sanfilippo, R. Rajesh, **S. Wheeler**, Z. Kilhoffer, & K. Zhou. "Commentary on Managing Misuse Risk for Dual-Use Foundation Models." *U.S. AI Safety Institute*, 2024. https://hdl.handle.net/2142/124759

Teaching Experience

Head Course Assistant, STAT 207: Data Science Exploration May 2026 – Present
University of Illinois Urbana-Champaign Urbana-Champaign, IL

Instructors: Dr. Julie Deeke & Dr. Kit Clement

- Help lead instruction and course operations for 500+ students studying core data science concepts, including statistical analysis and machine learning
- Develop and revise laboratory exercises and homework assignments to reinforce statistical reasoning, programming, and applied data science concepts
- Create, maintain, and update Canvas course infrastructure, including assignments, instructional materials, course organization, and student-facing resources

Professional Experience

Data Science & Research Intern May 2026 – August 2026
84.51° (Kroger) Cincinnati, OH

- Architected a multi-agent AI system that consolidates dataset metadata from GitHub, Jira, and Confluence into a semantic layer for downstream AI applications, reducing per-dataset consolidation time by 64%
- Developed an evaluation framework in MLflow containing 40+ programmatic checks and three LLM-as-a-judge scorers to assess agent output quality and reliability

Data Analytics Intern, Investor Services May 2025 – August 2025
ADM Chicago, IL

- Developed automated OFAC sanctions screening using layered fuzzy-matching techniques, including Levenshtein and Jaro–Winkler similarity, to support regulatory compliance during customer onboarding
- Engineered a two-layer futures-price forecasting model combining random forest prediction with gradient-boosted residual correction, reducing mean absolute error by 71%

Congressional Intern June 2023 – August 2023
U.S. House of Representatives Peoria, IL

- Conducted legislative research on technology and public-policy issues for congressional staff and members
- Managed confidential constituent casework data and coordinated with federal and state agencies to resolve constituent cases

Leadership & Service

Artificial Intelligence / Machine Learning Fellow May 2026 – Present
Break Through Tech Remote

- Selected from a pool of 4,300+ applicants for a 12-month artificial intelligence and machine learning fellowship

- Building a financial-resilience prediction system using Federal Reserve SHED microdata, comparing logistic regression, tree-based, and gradient-boosted models with probability calibration and SHAP-based explanations

Internal Operations Director May 2025 – January 2026
Illinois Business Consulting Champaign, IL

- Led internal project management and organizational strategy across marketing, alumni, and media teams, coordinating organization-wide engagement for 300+ members

Alumni Operations & Engagement Coordinator May 2024 – May 2025
Illinois Business Consulting Champaign, IL

- Built and launched the organization's first structured alumni engagement strategy, introducing a quarterly newsletter, annual alumni panel, and professional mentorship program

Selected Awards & Honors

Outstanding Undergraduate Poster Award, Illinois Undergraduate Research Symposium	2026
James Scholar Honors Program	2023 – Present
Dean's List, all semesters	2023 – Present
P.E.O. STAR Scholarship	2023

Research Methods & Technical Skills

Research Methods: Survey Design, Mixed-Methods Research, Experimental Design, IRB Protocol Development, Qualitative Thematic Coding, Computational Linguistics, Statistical Analysis, Predictive Modeling, Clustering

AI & Data: Machine Learning, Natural Language Processing, LLM Evaluation

Programming: Python, R, SQL, Java

Tools & Frameworks: Git, GitHub, Databricks, Google ADK, MLflow, Coh-Metrix, NLTK, TensorFlow, PyTorch

Visualization: Matplotlib, Seaborn, Plotly, Altair, Vega-Lite